

The effect of alcohol-specific episodic memories and alcohol cue exposure on value-based decision-making and ad libitum alcohol consumption

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Word count (excluding abstract, references, tables, and figures): 4286

Declarations of competing interest: Authors declare that they have no conflicts of interest.

Primary funding: This research was funded by the UK Medical Research Council (MR/W028476/1).

ABSTRACT

Background and Aims: Alcohol choice is governed by its subjective value, which can be experimentally modified. However, less is known about the moment-to-moment dynamics through which experimental manipulations of value influence observable choice behaviour. We tested whether an experimental manipulation intended to elevate alcohol value would alter computational parameters of value-based decision-making (VBDM) and alcohol choice, and whether these outcomes would mediate *ad libitum* alcohol consumption.

Design: Pre-registered, randomised between-subjects experiment comparing two personalised experimental conditions: alcohol value (exposure to alcohol memories and multisensory alcohol cues) vs. control (exposure to non-alcohol memories and multisensory soft-drink cues), conducted over two separate sessions with baseline measures included as covariates.

Setting: University laboratory in the United Kingdom (UK).

Participants: 129 adults (63% male; mean age = 44 years; mean Alcohol Use Disorders Identification Test [AUDIT] score = 16) who regularly consume alcohol (mean Estimated Weekly Alcohol Consumption [EWAC] = 42 UK units/week), randomised to alcohol value (n = 65) or control (n = 64) conditions.

Measurements: At baseline and post-exposure, participants completed self-reported measures of alcohol craving and a value-based decision-making (VBDM) task where they chose between pairs of previously rated alcohol-related and non-alcohol images. A drift-diffusion model was fitted to behavioural data to estimate evidence accumulation rates (value sensitivity) and response thresholds (response caution). Participants also completed a concurrent choice task (alcohol vs. non-alcohol) that measured percentage alcohol relative to

non-alcohol reward (enlargement; index of relative alcohol value), and a taste test to measure alcohol consumption.

Findings: Some indices of subjective craving were responsive to the experimental manipulation, but there were no significant between-group differences at post-manipulation ($p \geq .17$). Baseline-adjusted analyses showed that response thresholds for non-alcohol decisions were higher in the alcohol value condition compared to control ($p = .03$, $d = 0.40$); however, Bayesian analysis quantified only anecdotal evidence for this effect ($BF_{10} = 1.84$). Other VBDM parameters, percentage alcohol choice, and alcohol consumed were unaffected by the manipulation ($p \geq .19$, $BF_{10s} \leq 0.40$). Furthermore, no value indices (computational or percentage choice) mediated the effect of the experimental manipulation on alcohol consumed.

Conclusions: A personalised experimental manipulation intended to elevate alcohol value had modest effects on self-reported craving, but did not alter computational parameters of VBDM, relative alcohol choice, or ad-libitum alcohol consumption.

Keywords: Alcohol; Computational; Cue reactivity; Experimental; Memory; Value

Word count: 375 (max. 400)

INTRODUCTION

Many models of addiction conceptualise alcohol and other substance use as a dynamic, state-dependent behaviour shaped by contextual and environmental influences [1]. Bornstein and Pickard's memory-sampling framework [2] proposes that decisions about whether to drink alcohol (or not) are biased by selective sampling of highly rewarding episodic memories of past alcohol use. Viewed through the lens of behavioural economics [3], this memory sampling process influences the momentary subjective value of alcohol relative to available alternatives reinforcers. However, behavioural economic accounts and techniques are largely agnostic with respect to the specific cognitive mechanisms that underlie momentary choice [4].

Value-based decision-making (VBDM) provides a framework that can be used to quantify the specific cognitive mechanisms through which value is translated into discrete choice behaviour [5–7]. According to VBDM accounts [5], momentary decisions are guided by the overall value assigned to available response options, which is computed as the weighted sum of diverse value input signals comprising the anticipated gains (e.g., social reward, positive mood induction) and costs (e.g., effort, monetary price, potential adverse health/social consequences) associated with each response option. The highest-value option is selected through a value-to-action evidence accumulation (EA) process, which can be estimated from choices and reaction times in value-based choice tasks using the drift-diffusion model (DDM) [8]. This yields key decision parameters, including EA rate (the average rate at which evidence accumulates; interpreted as an index of sensitivity to the value of a given choice option [4]) and response threshold (response caution indexed by speed-accuracy trade-offs [9]). A core assumption of the DDM is that noisy (i.e., stochastically fluctuating) evidence accumulates until it reaches a threshold for responding, at which point

the decision is made (see [10] for a review). In the context of VBDM, this evidence refers to momentary information favouring one option over another based on their subjective value.

Prior addiction research has examined whether drug valuation processes are sensitive to experimentally induced motivational states [11,12]. A commonly used experimental tool is the concurrent choice task, in which participants repeatedly choose between images from two categories (e.g., alcohol versus snacks) to obtain a reward, such as points or enlargement of the preferred image [13]. In this task, relative choice of the drug over the alternative reward indicates subjective value of the drug. Using this paradigm, several studies have demonstrated that drug value is sensitive to experimental manipulation, including negative mood induction, deprivation, satiety, exposure to health warnings, imposition of time and delay costs, and taste aversion manipulations [12,14–18]. However, a limitation of concurrent choice tasks is that, although their sensitivity to relative value is consistent with molar theories of choice [19], they cannot disambiguate whether an experimentally induced change reflects a shift in value of the drug, the alternative reward value, or both.

Manipulations have also adopted multisensory and idiographic approaches to elicit personally salient motivational states, including preference-specific cues and personalised imagery scripts constructed from participants' descriptions [20–24]. In the context of cocaine use, personalised imagery scripts based on past cocaine-use experiences elicited moderate-to-intense craving in cocaine-dependent men [25]. Moving beyond the laboratory, Zacher et al. [26] examined spontaneous memories of past cocaine use in everyday life and found that more vivid or intrusive memories were associated with stronger craving. However, the extent to which these transient motivational states alter moment-to-moment decision processes that determine observable choice behaviour in addiction research remains largely unexplored.

More recently, addiction researchers have adopted VBDM tasks that overcome the limitations of standard behavioural measures of choice [27]. These tasks typically require participants to evaluate a set of images before choosing between two options from the same commodity category (e.g., alcohol, food) across multiple trials, generating both reaction time and choice data that can be modelled using computational approaches such as the DDM. This enables valuation processes to be estimated separately for each commodity (e.g., alcohol and non-alcohol commodities). Using this VBDM approach, experimental manipulations such as negative mood induction and consequence-focused framing of alcohol outcomes have been shown to influence valuation as captured by evidence accumulation rates. Importantly, these initial findings [28,29] suggest that these effects are driven by alterations in the value ascribed to alternatives to alcohol, rather than the value of alcohol itself. This distinction not only speaks to different intervention targets [30] but also reinforces concerns about the challenges of interpreting findings from the concurrent choice task, as noted above.

To date, no research has examined the combined influence of personalised alcohol cues and imagery scripts on VBDM for alcohol and alternative cues, or on subsequent alcohol consumption. This gap is notable given basic science accounts suggesting that recalled memories can directly feed into the evidence accumulation process that drive value-based choice [2,31]. The present study combines personalised, multisensory experimental manipulations with advances in computational methods for addiction research [27]. Primary hypotheses (pre-registered: <https://aspredicted.org/3bnq-yqpc.pdf>) were that, compared with a control condition, participants exposed to positive episodic alcohol memory combined with alcohol cue exposure (alcohol value condition) would show:

1. Higher EA rates and lower response thresholds when making alcohol-related decisions, and lower EA rates and higher response thresholds when making non-alcohol decisions (VBDM task).

2. Higher percentage choice of alcohol over the non-alcohol alternative reward (CC task), with this group difference mediated by VBDM parameters.
3. Higher volume of alcohol consumed, with this group difference mediated by VBDM parameters, over and above percentage alcohol choice from the CC task.

METHOD

Study design and transparency

Between-subjects randomised experiment with two conditions (alcohol value vs. control). Participants attended two in-person lab sessions (baseline and experimental); at baseline they recalled one alcohol-related and one non-alcohol-related autobiographical memory and reported their drink preferences. Following randomisation, condition-relevant memory and drink information were used in the experimental session. The study was approved by the University of Sheffield research ethics committee (Reference: 050755).

Study design, hypotheses, sample size, modelling approach, and statistical analyses plan were pre-registered before data was collected. Our pre-registration, anonymised data, and analysis scripts are available on ResearchBox (<https://researchbox.org/5984>)¹.

Participants

Informed by an *a priori* power analysis ($f = 0.35$, $\alpha = .01$, 90% power), we recruited 129 participants from the community in Sheffield using printed and online advertisements. All participants were ≥ 21 years old (mean age = 44.72, SD = 17.23) and consumed ≥ 28 UK units of alcohol per week (1 UK unit = 8g alcohol) which is at least double the current UK low-risk guidelines [32]. Exclusion criteria included: use of medication contraindicated with alcohol (e.g., benzodiazepines), any health condition affected by alcohol consumption, or

¹ The access code for these anonymised, peer-review materials is BHMURB.

prior treatment for alcohol use disorder or medical advice to seek such treatment. Participants provided informed consent and received a £50 Love2shop voucher. Recruitment occurred between April 2023 and April 2024.

The sample was 63% male and predominantly White (95%). Most participants were higher-risk drinkers, with 50% classified as hazardous drinkers, 23% as harmful drinkers, and 24% meeting criteria for probable alcohol dependence based on the Alcohol Use Disorders Identification Test (AUDIT) [33]. See Table S1 in the supplementary file for a detailed breakdown of participant characteristics.

Materials and measures

Questionnaires

Participants completed a battery of questionnaires, including the 10-item AUDIT [33] with extended AUDIT-C response options [34] (McDonald's $\omega = .71$); the 11-item Alcohol Symptom Checklist [35] ($\omega = .73$); the 14-item Alcohol Purchase Task (APT) [36]; the 12-item Drinking Motives Questionnaire–Revised Short Form (DMQ-SF) [37] (all subscales $\omega \geq .75$ except enhancement, $\omega = .58$); the 14-item Approach and Avoidance of Alcohol Questionnaire, Right Now version (AAAQ) [38] (all subscales $\omega \geq .78$); the 9-item Drinking Refusal Self-Efficacy Questionnaire [39,40] (all subscales $\omega \geq .73$); the 17-item Activity Level Questionnaire (ALQ) modified from [41]; the 8-item retrospective-memory subscale of the Prospective and Retrospective Memory Questionnaire (PRMQ) [42] ($\omega = .77$); the Positive and Negative Affect Schedule–Expanded (PANAS-X) [43] (all subscales $\omega \geq .78$); a single-item craving visual analogue scale assessing current urge to drink alcohol (0 “not at all” to 100 “a lot” slider); and demographic questions.

Personalised memory scripts and drink preferences

During the baseline session, participants recalled two positive autobiographical memories (one alcohol-related and one unrelated to alcohol; order randomised with a brief distractor in-between), first visualising and then describing each in detail. The researcher probed for episodic and vivid features (e.g., time, place, emotions, bodily sensations). Memories were audio-recorded and condensed into ~1-minute scripts post-session using recordings and notes to retain salient vivid/episodic features; the memory selected for summarisation depended on experimental condition (alcohol value vs. control). Twenty percent of the memory summaries were independently coded by two researchers, yielding 73.39% inter-rater agreement. Scripts were voice-recorded (by a gender-matched researcher where possible). Mood was assessed after each recall² using the PANAS-X [43]. We assessed alcohol and soft drink preferences by asking participants to rank their favourite drinks from a predefined list of commonly consumed beverages available in the UK (see supplementary file for details).

Pictorial stimuli

The 35 alcohol and soft drink images were sourced from the Amsterdam Beverage Picture Set [44] selecting common UK brands, supplemented with Google images to increase variety of recognisable options. The 35 images depicting alcohol-free reinforcement (e.g., exercise, family time), informed by previous research on substance-free reinforcement [45] and our own preliminary patient and public involvement consultations [46], were sourced from copyright-free databases (e.g., <https://www.pexels.com/>)³.

² Across participants, sadness scores were higher, and joviality scores lower, following alcohol compared to non-alcohol memory retrieval ($ps < .001$; $ds \geq .31$).

³ Primary pre-registered analyses focused on alcohol versus alcohol-free reinforcement images; therefore, the term ‘non-alcohol’ is used throughout the reporting of results in the manuscript to refer specifically to the alcohol-free reinforcement condition. Supplementary analyses additionally examined soft-drink images as an appetitive alternative comparator to alcohol.

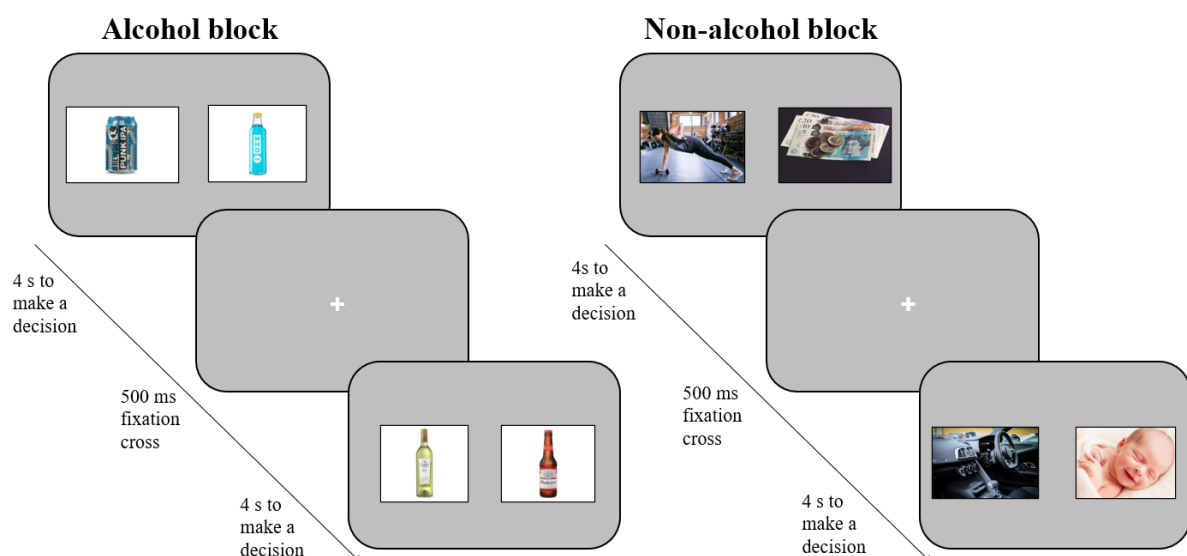
Image-rating phase

Participants sorted the 35 images from each set (alcohol, soft drink, and alcohol-free reinforcement) into four value categories ('A lot', 'A little bit', 'Not really', 'Not at all') via drag-and-drop, assigning at least five per category. From each category, five images were then randomly selected for use in the subsequent VBDM task.

VBDM task [27]

On each trial (Figure 1), two images of the same set appeared in the centre of the screen, and participants were instructed to use one of two keys to choose the image that they liked more ('Z' for left and 'M' for right) as quickly as possible. Following practice trials, participants completed three randomised blocks (alcohol, soft drink, alcohol-free reinforcement) of 150 trials (450 trials in total), with breaks every 50 trials. Trial difficulty was determined by the rating difference between paired images from the image-rating phase (1 = hard, 2 = medium, 3 = easy). On each trial there was a "correct" answer (i.e., preference consistent with previous rating); whether this appeared on the left or the right was randomised. Responses > 4s were classed as "miss trials" [47].

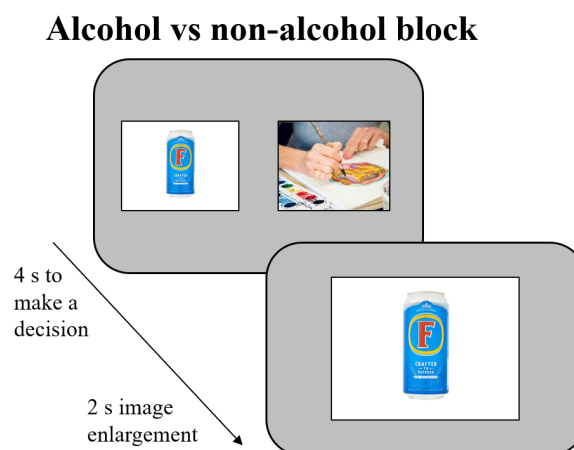
Figure 1. *Example trials in the value-based decision-making (VBDM) task*



Concurrent choice task [48]

Only the five highest-rated images per set were used (five per set; 15 total). On each trial (Figure 2), an alcohol image was paired with either an alcohol-free reinforcement or soft drink image, presented centrally. Participants used a computer key ('Z' for left and 'M' for right) to select an image to enlarge. Following practice trials, participants completed two randomised blocks (alcohol vs. alcohol-free reinforcement; alcohol vs. soft drink) of 50 trials each (100 trials in total), with a break after 50 trials. Whether the alcohol image appeared on the left or the right was randomised.

Figure 2. *Example trials in the concurrent choice (CC) task*



Taste test [49]

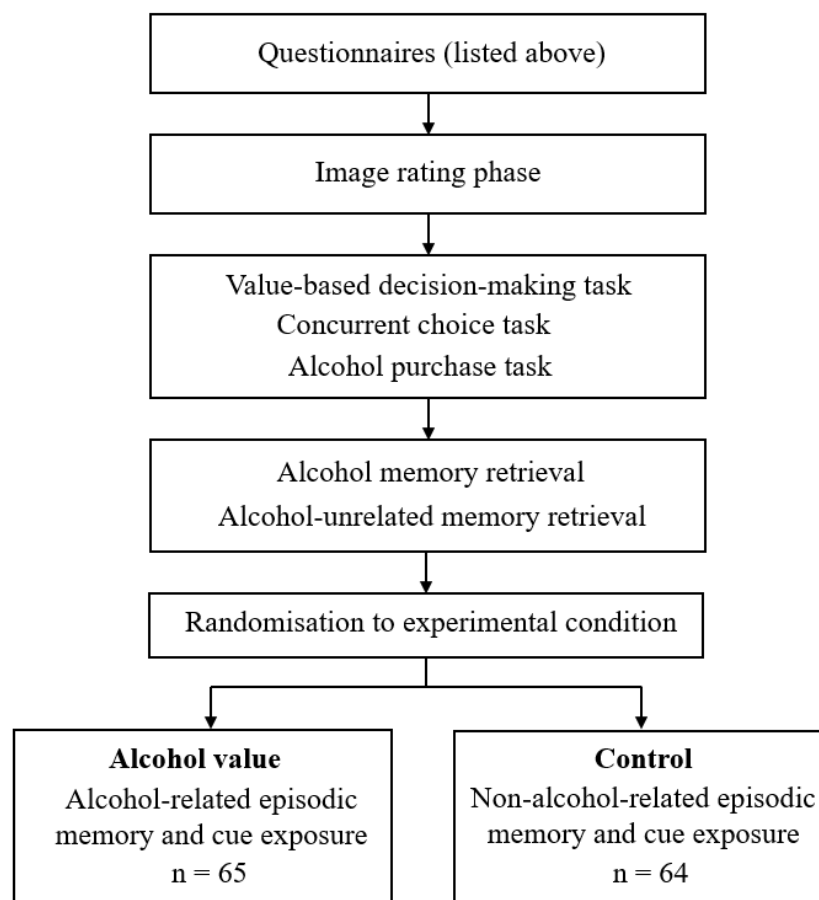
Ad libitum alcohol consumption was assessed using a bogus taste test. Participants received 1 UK unit of each of their two preferred alcoholic drinks (2 units total), rated them on ten dimensions (e.g., light, fizzy, pleasant), and answered comparative questions about the drinks. Before the test, they rated thirst (100-mm VAS) and were told alcohol might impair their performance on a subsequent cognitive task carrying a potential £5 reward (a deception; all received £5). This instruction was intended to introduce a perceived cost to drinking,

thereby encouraging variability in consumption. Participants were left alone for 15 minutes and could drink as much or as little as desired.

Procedure

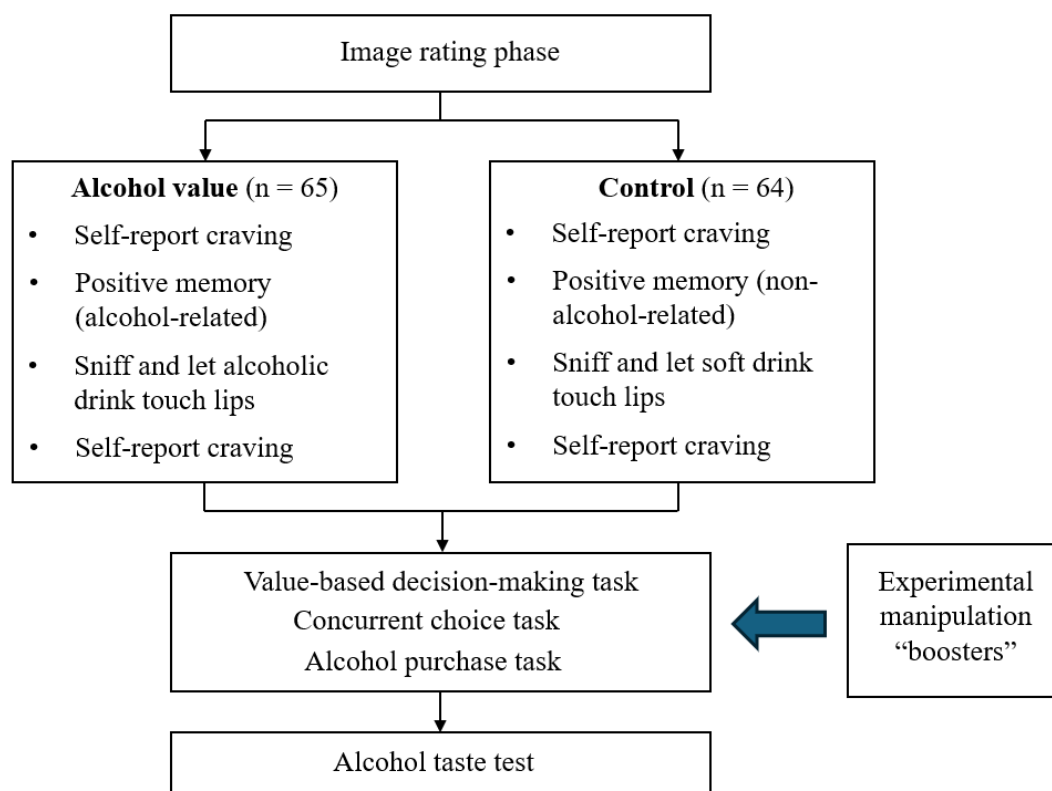
Interested participants firstly completed a 15-minute telephone screen to assess eligibility. Those eligible were subsequently invited to attend two laboratory sessions on the University of Sheffield campus (baseline and experimental), scheduled where possible within 10 days of each other ($M = 8.30$ days, $SD = 4.74$; range = 1–35 days). Each session lasted approximately 1.5 hours (3 hours total). Sessions took place between 12:00pm and 9:00pm, and participants provided a 0.00% breath alcohol concentration (BAC) reading before each session. See Figures 3 and 4 for an overview of the study procedure.

Figure 3. *Schematic overview of session 1 (baseline)*



Immediately after the baseline session, participants were randomly assigned (1:1) to the increased alcohol value condition ($n = 65$) or control condition ($n = 64$) using a pre-generated computer-randomised allocation sequence. Researchers were blinded to condition assignment until after participants completed the baseline session, while participants remained blinded until they arrived for the second session. Condition assignment determined the materials researchers prepared for the subsequent experimental session.

Figure 4. *Schematic overview of session 2 (experimental manipulation)*



In the alcohol value condition, participants listened via headphones to a recorded summary of their alcohol-related memory while imagining themselves in the scene with their eyes closed. Imagery continued until instructed to stop (30 s after the script ended). Before and after memory playback, participants were presented with their preferred alcoholic drink and instructed to sniff it, touch it to their lips three times without drinking, and consider how consuming it would make them feel. The control condition procedure was identical, except that participants listened to their non-alcohol-related memory and were presented with their

preferred soft drink. All participants completed the AAAQ and a single-item craving measure before and after the experimental manipulation. They then completed the decision-making tasks (VBDM and CC; in that order) followed by the APT. During task breaks, participants repeated the sniffing and lip-touching procedure with their assigned drink as manipulation “boosters”. The session concluded with the taste test. Finally, participants were debriefed and completed a short cool-down exercise to minimise residual craving (see supplementary materials).

Data preparation and analysis

On the VBDM task, miss ($> 4s$; 0.64%) and fast response ($< 300ms$; 0.04%) trials were excluded (0.68% total). The DDM [8] was then fitted to accuracy and response time data using the EZ method [50]. For the CC task, miss ($> 4s$; 0.46%) and fast response ($< 300ms$; 0.09%) trials were excluded (0.55% total).

Analyses of covariance (ANCOVAs) were conducted on post-manipulation VBDM parameters (EA rates and response thresholds) and percentage alcohol choice, with condition (alcohol value vs. control) as a between-subjects factor and the corresponding pre-manipulation (baseline) value included as a covariate. Statistical analyses were conducted in RStudio (version 4.5.1) [51]. To aid interpretation of results, we calculated Bayes factors in JASP (version 0.19.2) [52] using default priors, with values > 3 and < 0.3 indicating evidence for the experimental and null hypotheses, respectively [53].

RESULTS

Descriptive statistics for the self-report measures are presented in Table 1.

Table 1. *Scores on self-report questionnaires (values are mean and standard deviation), split by experimental condition*

	Alcohol value (<i>n</i> = 65)	Control (<i>n</i> = 64)	<i>p</i> -value and effect size
AUDIT score	16.83 (5.94)	16.06 (4.67)	<i>p</i> = .42, <i>d</i> = 0.14
EWAC score	42.10 (20.29)	42.45 (24.47)	<i>p</i> = .93, <i>d</i> = 0.02
Self-efficacy (social)	2.55 (1.46)	2.49 (1.33)	<i>p</i> = .81, <i>d</i> = 0.04
Self-efficacy (emotional)	4.29 (1.94)	3.97 (1.70)	<i>p</i> = .32, <i>d</i> = 0.17
Self-efficacy (opportunity)	5.60 (1.35)	5.22 (1.42)	<i>p</i> = .12, <i>d</i> = 0.27
Coping motives	2.42 (1.13)	2.51 (1.08)	<i>p</i> = .65, <i>d</i> = 0.08
Conformity motives	1.86 (0.95)	1.84 (0.85)	<i>p</i> = .94, <i>d</i> = 0.01
Social motives	3.66 (1.08)	3.60 (0.93)	<i>p</i> = .73, <i>d</i> = 0.06
Enhancement motives	3.57 (0.82)	3.41 (0.79)	<i>p</i> = .24, <i>d</i> = 0.21
Alcohol-related reinforcement	3.13 (1.50)	3.12 (1.54)	<i>p</i> = .97, <i>d</i> = 0.01
Alcohol-free reinforcement	5.97 (2.04)	6.17 (2.10)	<i>p</i> = .68, <i>r</i> _{rb} = 0.04
Reinforcement ratio	0.34 (0.12)	0.33 (0.11)	<i>p</i> = .63, <i>d</i> = 0.09
Retrospective memory ability	19.45 (4.61)	18.84 (4.69)	<i>p</i> = .46, <i>d</i> = 0.13
Years since alcohol memory	9.21 (11.66)	10.14 (13.60)	<i>p</i> = .68, <i>d</i> = 0.07
Years since non-alcohol memory	8.13 (11.85)	7.71 (12.51)	<i>p</i> = .48, <i>r</i> _{rb} = 0.07
Joviality post alcohol memory	18.85 (7.20)	17.70 (6.54)	<i>p</i> = .35, <i>d</i> = 0.17
Joviality post non-alcohol memory	21.35 (6.74)	18.77 (7.26)	<i>p</i> = .04, <i>d</i> = 0.37
Sadness post alcohol memory	1.60 (3.05)	1.59 (2.61)	<i>p</i> = .33, <i>r</i> _{rb} = 0.09
Sadness post non-alcohol memory	0.77 (1.58)	1.19 (2.26)	<i>p</i> = .10, <i>r</i> _{rb} = 0.14

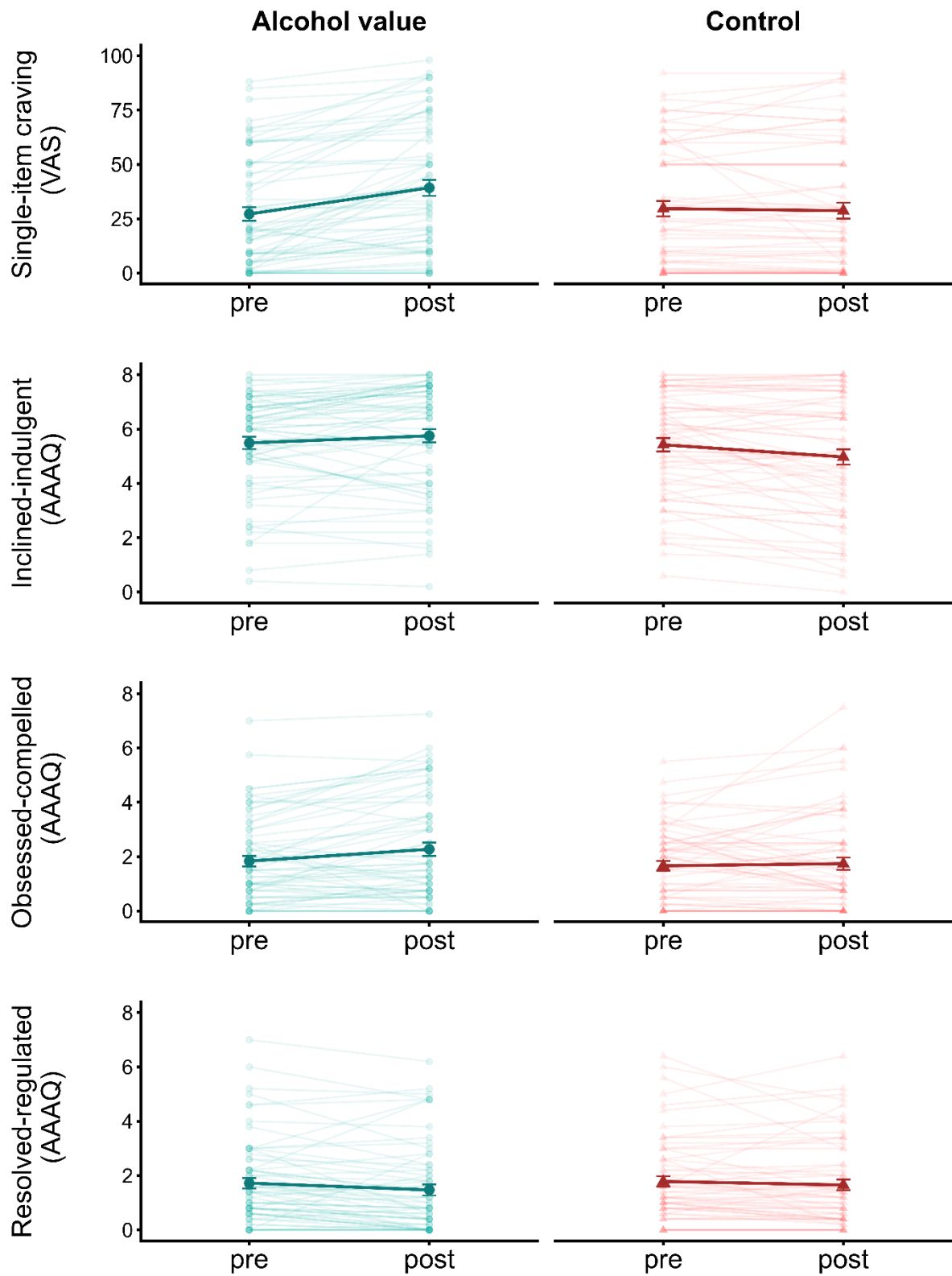
Note. Effect sizes are Cohen's *d* (for data that are approximately normally distributed) or rank-biserial correlations (*r*_{rb}) (for data that are not approximately normally distributed).

Effect of experimental manipulation on craving

To examine the effect of the experimental manipulation on self-reported alcohol craving, single-item craving and AAAQ scores were analysed pre- and post-manipulation (Figure 5). Single-item craving was analysed using a two-way mixed ANOVA (time x condition), and AAAQ ratings were analysed using a three-way mixed ANOVA (subscale x time x condition). There was a significant time x condition interaction for single-item craving, $F(1, 127) = 33.43, p < .001, \eta^2 = .21$. Post-hoc comparisons (applying Bonferroni–Holm correction) showed that single-item craving increased in the alcohol value condition ($p < .001, d = 0.43$), but not in the control condition ($p = 1.00, d = 0.03$). Post-manipulation craving did not differ significantly between conditions ($p = .18, d = 0.37$).

There was a significant subscale x time x condition interaction for AAAQ scores, $F(1.77, 224.27) = 6.91, p = .002, \eta^2 = .05$. Follow-up two-way ANOVAs (time x condition) conducted separately for each subscale revealed a significant interaction for the inclined-indulgent subscale, $F(1, 127) = 16.78, p < .001, \eta^2 = .12$, but not for the obsessed-compelled, $F(1, 127) = 3.74, p = .06, \eta^2 = .03$, or resolved-regulated subscales, $F(1, 127) = 0.67, p = .42, \eta^2 = .01$. Post-hoc comparisons showed that obsessed-compelled scores increased in the alcohol value condition ($p = .006, d = 0.26$), but not in the control condition ($p = 1.00, d = 0.05$). Inclined-indulgent scores decreased in the control condition ($p = .002, d = 0.22$), with no significant change in the alcohol value condition ($p = .14, d = 0.13$). No significant changes were observed for the resolved-regulated subscale in either the alcohol value ($p = .11, d = 0.16$) or control condition ($p = 1.00, d = 0.08$). Post-manipulation comparisons indicated no significant between-condition differences for the inclined-indulgent ($p = .17, d = 0.39$), obsessed-compelled ($p = .44, d = 0.31$), or resolved-regulated ($p = 1.00, d = 0.12$) subscales.

Figure 5. *Self-reported craving and AAAQ scores pre- and post-experimental manipulation in the alcohol value (left) and control (right) conditions*



Note. VAS = visual analogue scale; AAAQ = Approach and Avoidance of Alcohol

Questionnaire. Translucent lines represent individual participant trajectories.

Pre-registered analyses

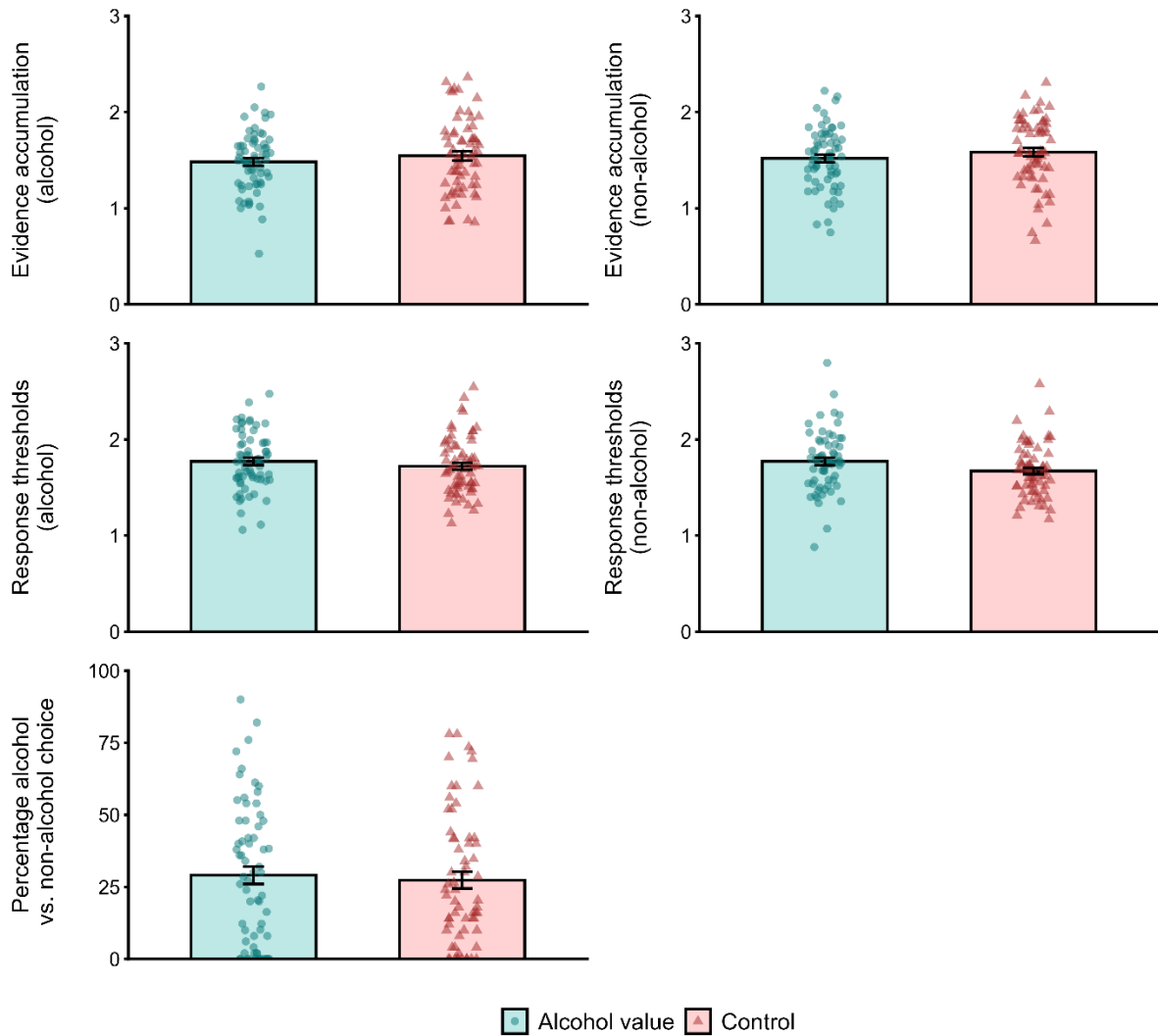
Hypothesis 1

Controlling for baseline values, the effect of condition on non-alcohol response thresholds was significant, $F(1, 126) = 5.07, p = .03, \eta^2 = .04$; however, this did not meet our corrected significance threshold ($\alpha = .01$), and Bayesian analyses indicated only anecdotal evidence for this effect ($BF_{10} = 1.84$). Consistent with our hypothesis, post-hoc comparisons revealed that non-alcohol response thresholds were higher in the increased alcohol value condition ($M = 1.77; SD = 0.31$) compared to the control condition ($M = 1.67; SD = 0.27; p = .03, d = 0.40$). There was no significant effect of condition on non-alcohol EA rates ($F(1, 126) = 0.44, p = .51, \eta^2 = .00, BF_{10} = 0.23$), alcohol EA rates ($F(1, 126) = 0.35, p = .56, \eta^2 = .00, BF_{10} = 0.22$), or alcohol response thresholds ($F(1, 126) = 0.02, p = .90, \eta^2 = .00, BF_{10} = 0.19$), with all Bayes factors indicative of moderate evidence in favour of the null. See Figure 6.

Hypothesis 2

There was no significant effect of the experimental manipulation on percentage alcohol choice ($F(1, 126) = 1.72, p = .19, \eta^2 = .01$) with Bayesian analysis indicating anecdotal evidence in favour of the null hypothesis ($BF_{10} = 0.40$). See Figure 6.

Figure 6. *Mean evidence accumulation (EA) rates and response thresholds for alcohol and non-alcohol decisions, and percentage alcohol choice, by experimental condition (alcohol value vs. control)*



Note. Error bars represent the standard error (SE) of the mean. Points show individual participant values (jittered). Analyses control for baseline values.

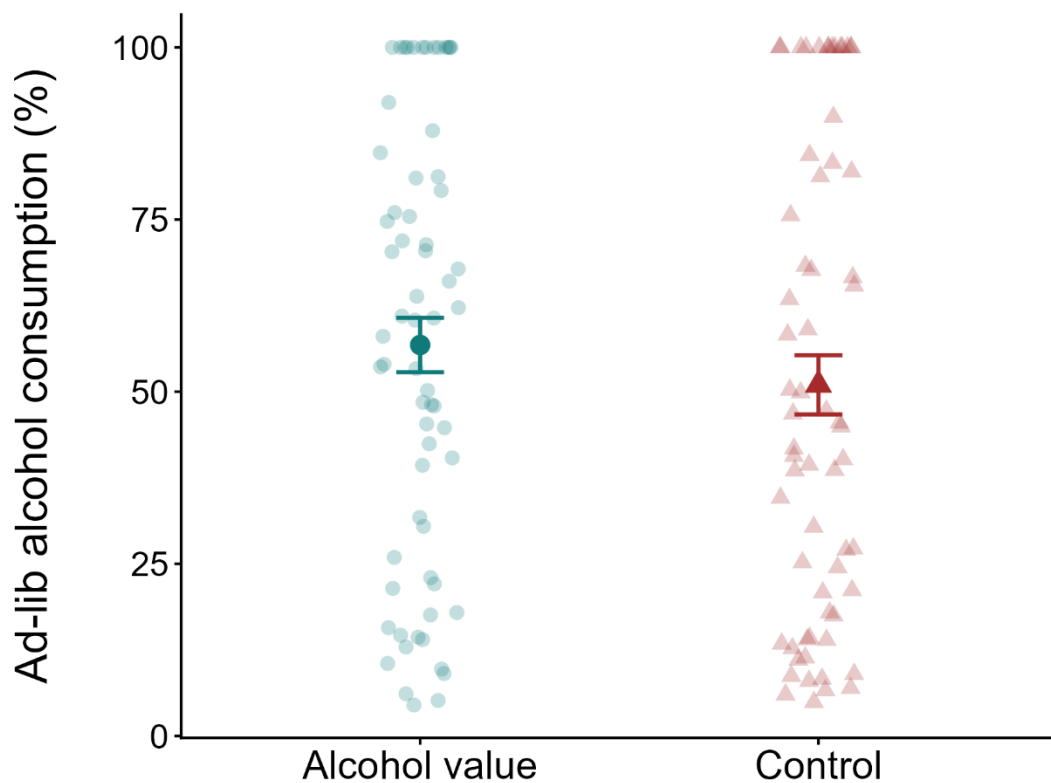
Although no significant effect of the experimental manipulation on percentage alcohol choice (CC task) was observed, we examined whether the association between experimental condition and percentage alcohol choice was indirectly explained by VBDM parameters. Baseline parameter values were included as covariates for their respective parameters. No statistically significant mediation effects were identified for any VBDM parameter (non-alcohol EA rates, $\beta = .01$, 95% CI = $-.01$ to $.05$; non-alcohol response thresholds, $\beta = .01$, 95% CI = $-.03$ to $.06$; alcohol EA rates, $\beta = .00$, 95% CI = $-.02$ to $.04$; alcohol response thresholds, $\beta = -.01$, 95% CI = $-.06$ to $.02$). The overall model explained approximately 20%

of the variance in percentage alcohol choice ($R^2 = .20$); however, some fit indices fell outside acceptable ranges, warranting cautious interpretation of the estimated pathways. Full results are provided in the Supplementary Materials (Figure S1).

Hypothesis 3

Participants in the alcohol value condition ($M = 56.78\%$, $SD = 31.87$) did not consume significantly more of their preferred alcohol than those in the control condition ($M = 51.00\%$, $SD = 34.40$), $t(127) = 0.99$, $p = .32$, $d = 0.17$. Bayesian analysis provided moderate evidence in favour of the null hypothesis ($BF_{10} = 0.29$).

Figure 7. Mean alcohol consumption, expressed as a percentage of the total amount offered, split by experimental condition



Note. Error bars represent the standard error (SE) of the mean. Points show individual participant values (jittered). Analyses control for participant thirst.

Although no significant effect of the experimental manipulation on alcohol consumption during the bogus taste test was observed (Figure 7), we examined whether the association between experimental condition and ad libitum alcohol consumption was indirectly explained by VBDM parameters. To determine whether VBDM parameters accounted for variance in alcohol consumption over and above that explained by percentage alcohol choice, this variable was included as a covariate in the model. In addition, baseline values for each VBDM parameter and participant thirst were included as covariates. No statistically significant mediation effects were identified for any VBDM parameter (non-alcohol EA rates, $\beta = -.01$, 95% CI = $-.06$ to $.01$; non-alcohol response thresholds, $\beta = .01$, 95% CI = $-.03$ to $.08$; alcohol EA rates, $\beta = .01$, 95% CI = $-.01$ to $.06$; alcohol response thresholds, $\beta = .00$, 95% CI = $-.04$ to $.01$). The overall model explained approximately 19% of the variance in alcohol consumption ($R^2 = .19$); however, some fit indices fell outside acceptable ranges, warranting cautious interpretation of the estimated pathways. Full results are provided in the Supplementary Materials (Figure S2).

Secondary analyses

Secondary analyses using soft drinks as an appetitive non-alcohol comparator and self-reported demand quantified by the Alcohol Purchase Task [36] revealed no effects consistent with the preregistered hypotheses (supplementary materials, Figures S3–S6).

DISCUSSION

Prior experimental research has shown that drug value is sensitive to a broad range of manipulations designed to alter motivation for drug consumption (see [13]), yet less is known about the moment-to-moment dynamics through which experimental manipulations of value influence observable choice behaviour. Drawing on VBDM and memory-sampling accounts of value-based choice [2,4,31,54], we found that the experimental manipulation intended to

elevate alcohol value did not robustly alter VBDM parameters, percentage alcohol choice, or ad-libitum alcohol consumption. Furthermore, there was no evidence in support of any mediating effects of VBDM parameters, either on percentage alcohol choice or ad-libitum alcohol consumption. The only indication of an effect was a higher response threshold for non-alcohol choices following exposure to positive alcohol memories and alcohol cues; however, this did not meet the corrected significance threshold ($\alpha = .01$), and Bayesian evidence for the alternative hypotheses was only anecdotal.

One potential explanation for the divergence from the broader VBDM literature is that the personalised, multisensory manipulation was not powerful enough to increase the momentary value of alcohol, potentially accounting for the modest effect on craving and the absence of reliable effects on VBDM parameters, concurrent choice, and ad libitum consumption. This is surprising given prior research findings showing that multisensory cue exposure and autobiographical recall can intensify craving [22,25,26]. However, there are important methodological differences: while the present study used a power calculation to determine a sufficient sample size, Kilts et al. [25] recruited only eight male participants, and their guided imagery scripts focused on anticipation rather than the reinforcing effects of cocaine use thereby limiting direct comparison with the present study. Beyond these design considerations, using ecological momentary assessment, Zacher et al. [26] found that 42.4% of cocaine-related memory intrusions occurred in the absence of craving, suggesting drug memories and craving can be dissociated.

Another important consideration is that autobiographical drinking episodes may have evoked mixed emotional responses that were not exclusively positive in nature. Indeed, two experimental studies found that prompting participants to recollect their most recent drinking occasion and estimate the calories consumed caused lower alcohol motivation relative to a neutral control condition, as indexed by both reduced craving [55] and slower rates of

beverage consumption [56]. Although the present study emphasised recollection of positive alcohol memories, it is possible that recalling autobiographical drinking experiences nonetheless elicited mixed or ambivalent emotional responses. In partial support of this, sadness scores were higher and joviality scores lower following alcohol compared to non-alcohol memory retrieval, suggesting that alcohol memory recall may have attenuated positive affect to some degree. It is also possible that the laboratory setting may also have constrained the emotional impact of the manipulation (see [57]). These considerations may account for the weak craving effects observed in the present study.

A further possibility is that the theoretical foundation for this study, grounded in value-based decision-making (VBDM) models [4,54], is weaker than anticipated, and that these models may require revision. Although findings from prior studies partially align with predictions derived from VBDM models [20,28,29,58,59], other studies did not provide support [60,61]. Considering the broader literature alongside the present findings, models of the role of value-based choice in addiction may require substantial modification or abandonment in favour of other models, although further work with more powerful experimental manipulations of drug value will be important.

A related but distinct avenue concerns the sensitivity of the VBDM task employed in the present study. It is possible that the task (in its current form) is not consistently sensitive to experimental manipulations, and that methodological refinement is needed. Recent iterations of the paradigm (e.g., [20]) have shifted from modelling absolute value to modelling relative value using more complex, custom DDM-fitting procedures.

This study has several strengths. Fitting the EZ-DDM to raw behavioural data has been found to enhance statistical power in group comparisons without requiring more participants or task trials [9]. Furthermore, personalising experimental materials to

participants' drink preferences and autobiographical memories enhanced ecological validity, while blinding researchers to condition allocation until after baseline assessment reduced the risk of bias. However, several limitations remain. First, ecological validity may have been limited because the controlled laboratory setting may not adequately simulate the situational contexts in which alcohol typically serves a valued function (e.g., emotional or social). Second, sessions were conducted from midday onwards which may not reflect typical (evening) drinking contexts. Third, the experimental manipulation of alcohol value yielded only weak effects on self-reported craving, suggesting that future research should explore more potent methods of elevating alcohol value to better capture its influence on choice and consumption. Indeed, the evidence for cue-reactivity at present is considered more mixed and contested than the field once assumed [62]. Finally, the sample was predominantly White (~95%), limiting generalisability to more ethnically diverse populations; future research should prioritise recruiting representative samples to establish broader applicability.

To conclude, a personalised experimental manipulation of alcohol value did not robustly alter computational parameters of VBDM, relative alcohol choice, or ad libitum alcohol consumption. Further research is required to determine if these results reflect a weak experimental manipulation, insensitive experimental tools, or if theoretical models of VBDM in addiction require refinement.

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